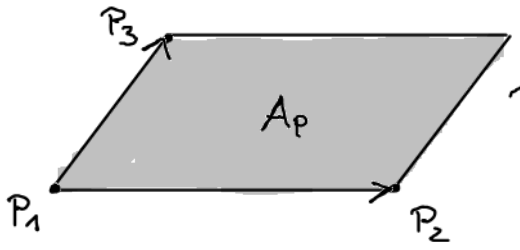
Lösung: Flächeninhalte

a)



– Parallelogramm

$$\overrightarrow{P_1P_3} = \overrightarrow{OP_3} - \overrightarrow{OP_1} = \begin{pmatrix}1\\-1\\2\end{pmatrix} - \begin{pmatrix}2\\3\\1\end{pmatrix} = \begin{pmatrix}-1\\-4\\1\end{pmatrix}$$

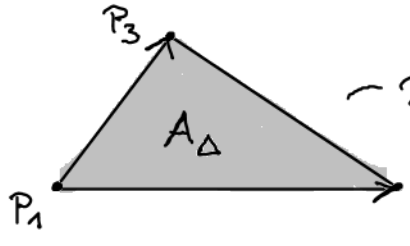
$$\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = \begin{pmatrix}-4\\1\\3\end{pmatrix} - \begin{pmatrix}2\\3\\1\end{pmatrix} = \begin{pmatrix}-6\\-2\\2\end{pmatrix}$$

$$A_P = \left|\overrightarrow{P_1P_3} \times \overrightarrow{P_1P_2}\right| = \left|\begin{pmatrix}-1\\-4\\1\end{pmatrix} \times \begin{pmatrix}-6\\-2\\2\end{pmatrix}\right|$$

$$= \left|\begin{pmatrix}(-4)\cdot 2 - 1\cdot(-2)\\1\cdot(-6) - 2\cdot(-1)\\(-1)(-2) - (-6)\cdot(-4)\end{pmatrix}\right| = \left|\begin{pmatrix}-6\\-4\\-22\end{pmatrix}\right| = \sqrt{(-6)^2 + (-4)^2 + (-22)^2}$$

$$= \sqrt{2^2\left((-3)^2 + (-2)^2 + (-11)^2\right)} = 2\cdot\sqrt{9+4+121} = \underline{\underline{2\cdot\sqrt{134}}}$$

b)



– Dreieck

$$A_\Delta = \frac{1}{2}\underbrace{\left|\overrightarrow{P_1P_3} \times \overrightarrow{P_1P_2}\right|}_{= A_P} = \frac{1}{2}\cdot\left(2\cdot\sqrt{134}\right) = \sqrt{134}$$